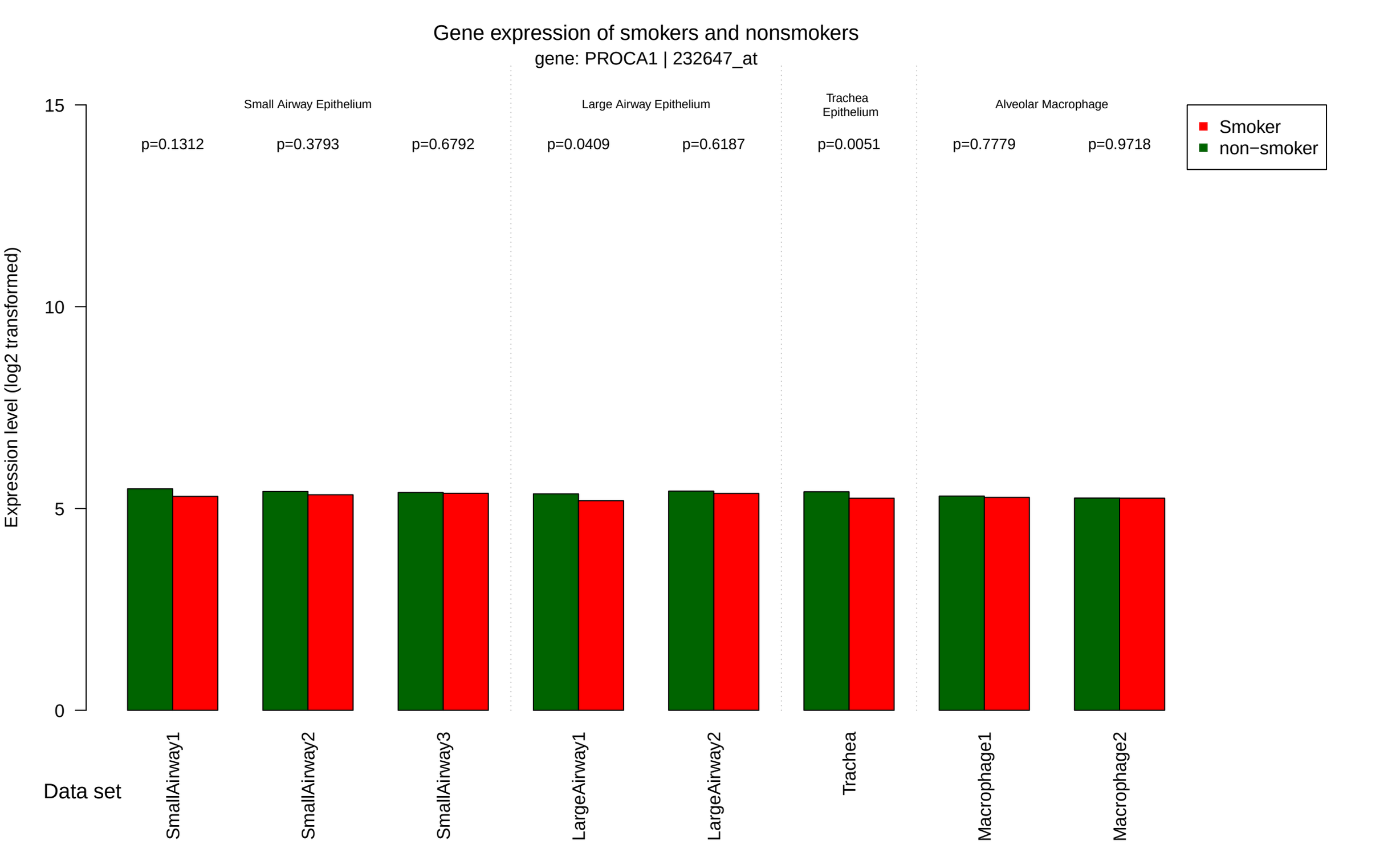
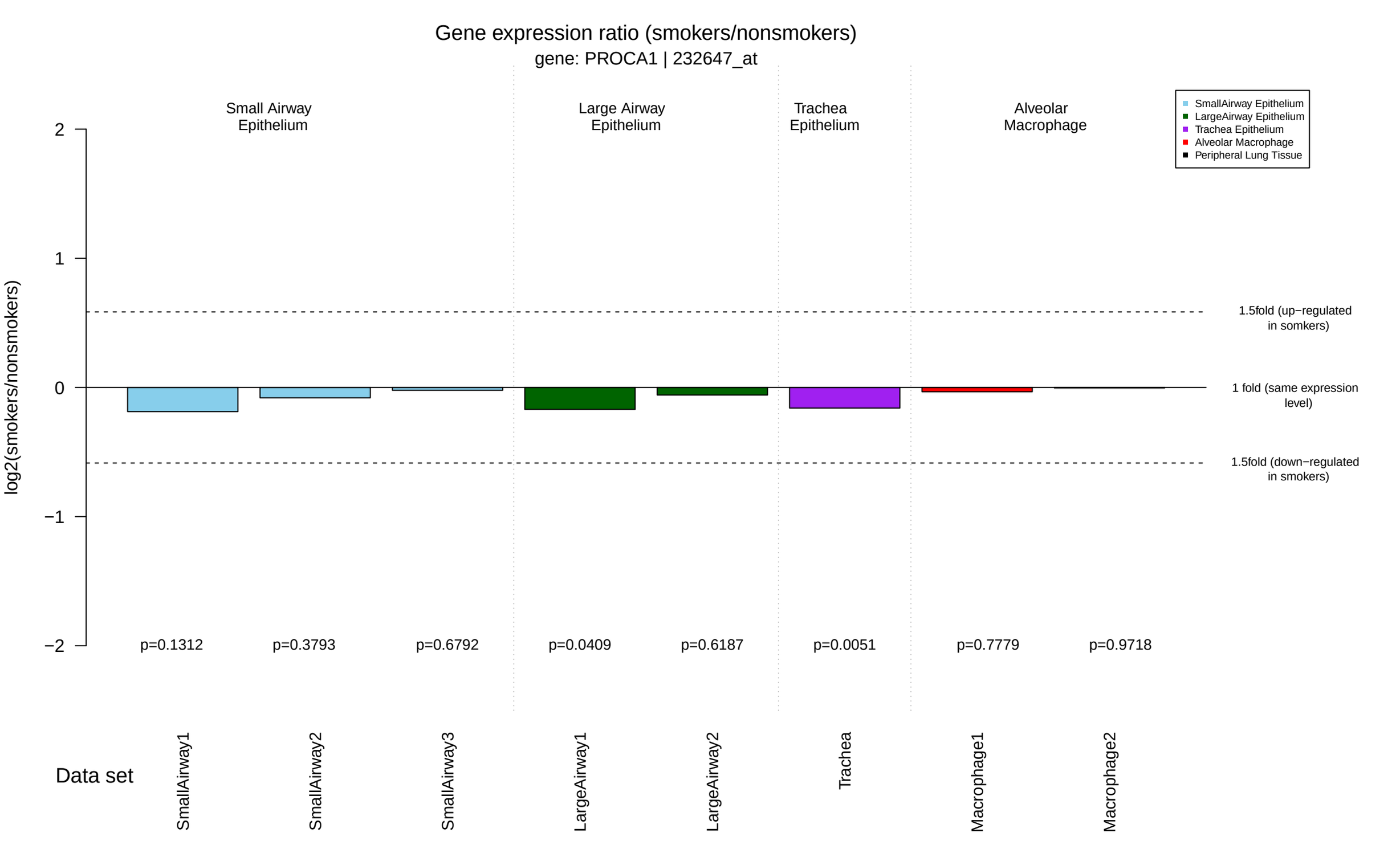
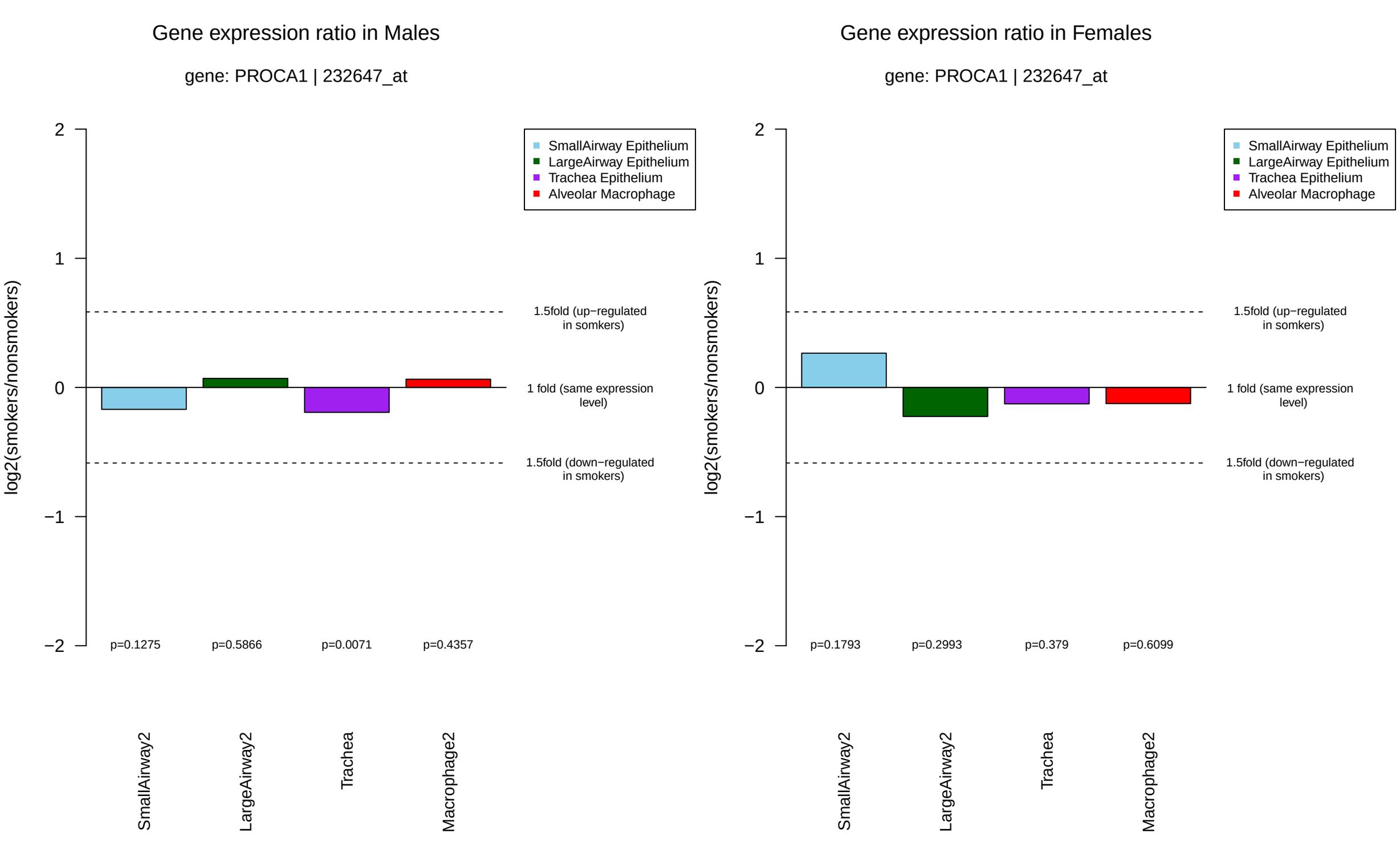
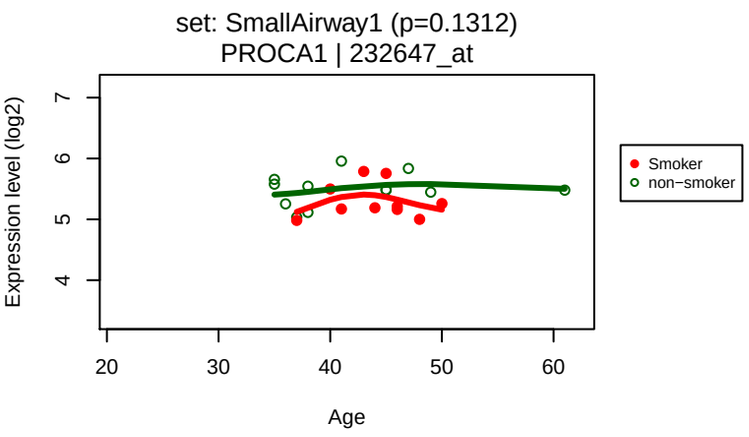
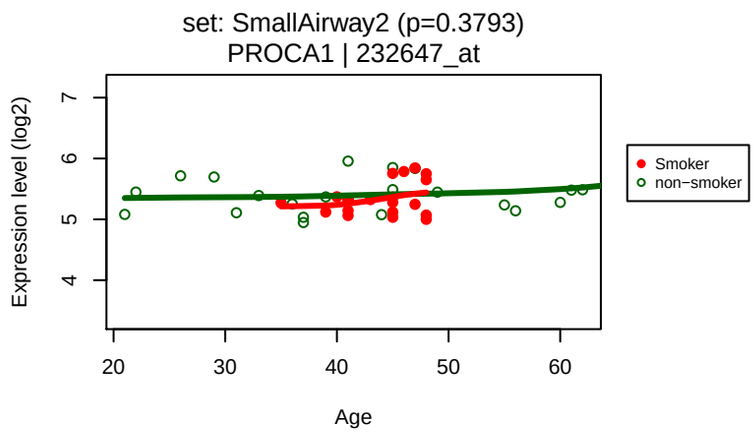
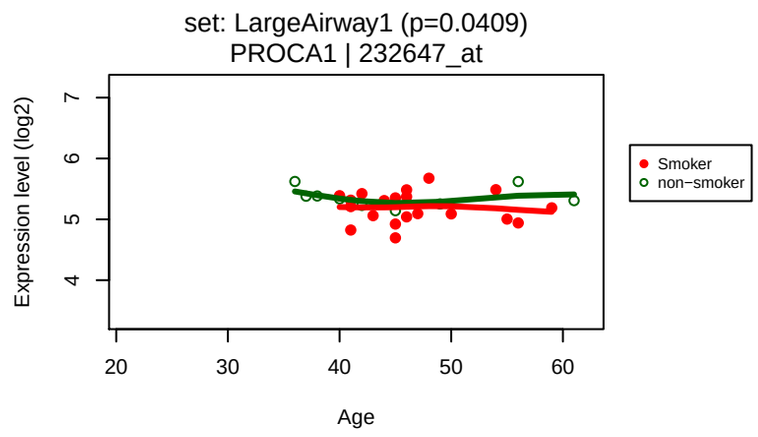

Gene expression of smokers and nonsmokers
gene: PROCA1 | 232647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1312
p=0.3793
p=0.6792
p=0.0409
p=0.6187
p=0.0051
p=0.7779
p=0.9718
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

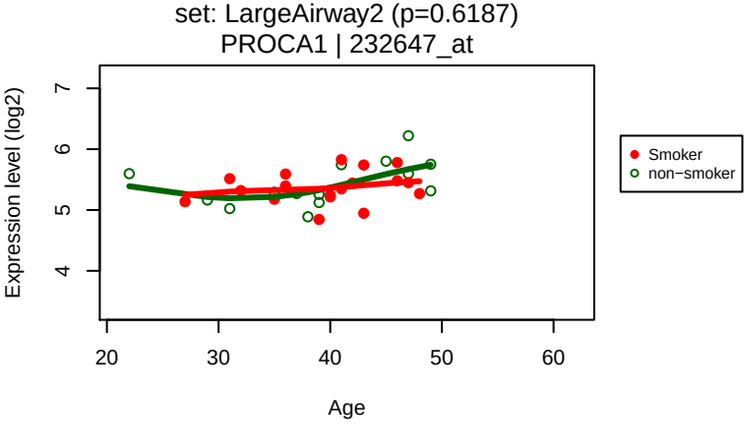

Gene expression ratio (smokers/nonsmokers)
gene: PROCA1 | 232647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1312
p=0.3793
p=0.6792
p=0.0409
p=0.6187
p=0.0051
p=0.7779
p=0.9718
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

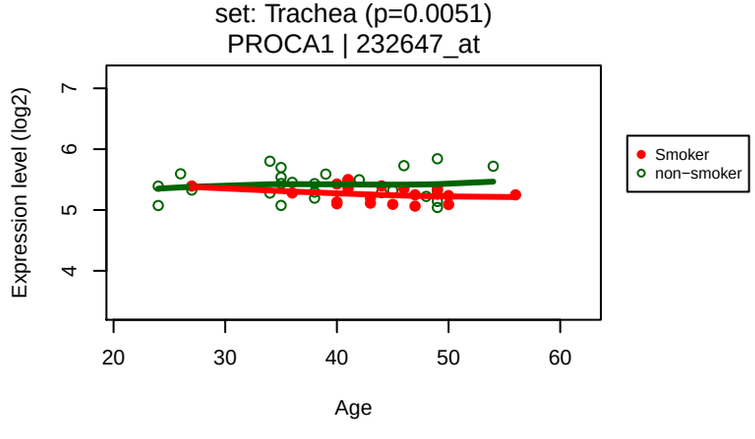

Gene expression ratio in Males
gene: PROCA1 | 232647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1275
p=0.5866
p=0.0071
p=0.4357
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PROCA1 | 232647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1793
p=0.2993
p=0.379
p=0.6099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

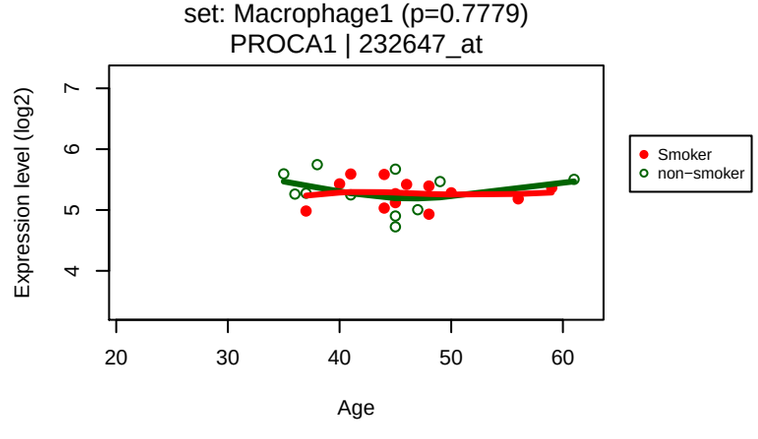

set: SmallAirway1 (p=0.1312)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker

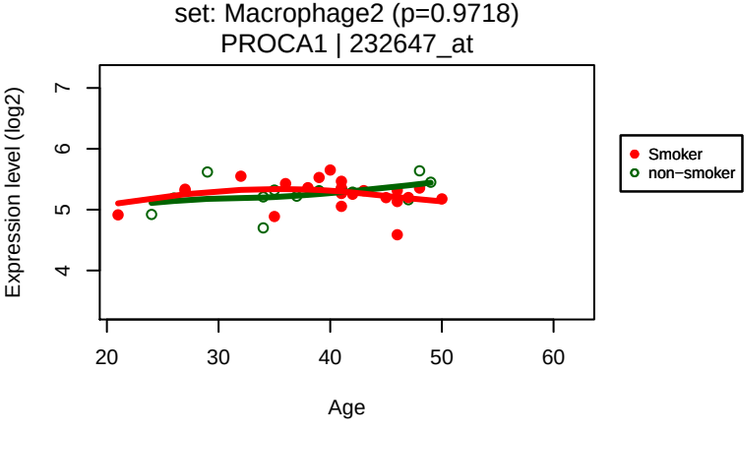

set: SmallAirway2 (p=0.3793)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker

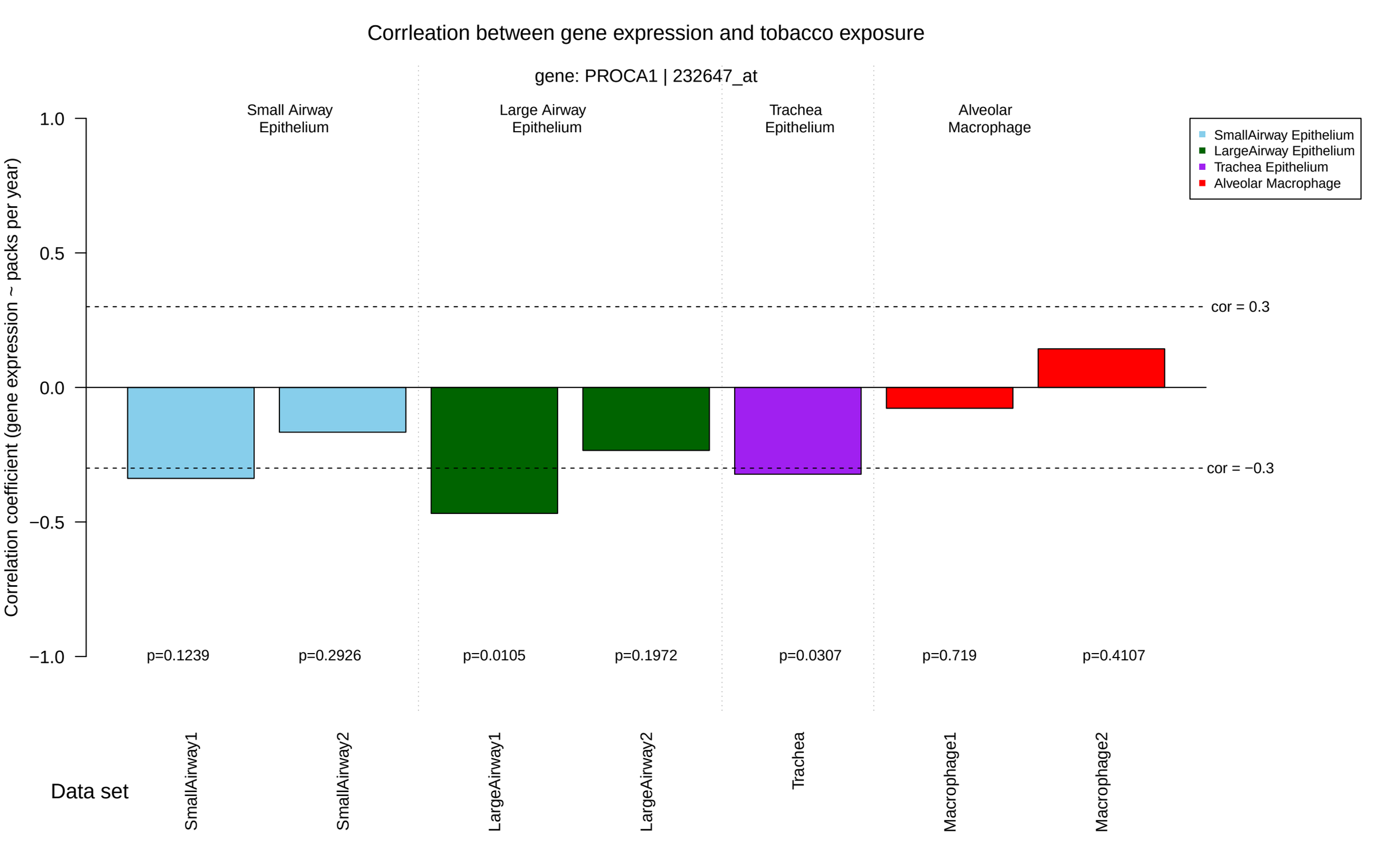

set: LargeAirway1 (p=0.0409)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6187)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7779)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9718)
PROCA1 | 232647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PROCA1 | 232647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1239
p=0.2926
p=0.0105
p=0.1972
p=0.0307
p=0.719
p=0.4107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2